EXPERIMENT NO. 1

ER Diagram for an Indian E-Commerce Platform

AIM

Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery and Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product and participation constraints.

ENTITIES AND PRIMARY KEYS

Customer – CustomerID (Primary Key)
Product – ProductID (Primary Key)
Order – OrderID (Primary Key)
OrderItem – OrderID + ProductID (Composite PK, Weak Entity)
Seller – SellerID (Primary Key)
Category – CategoryID (Primary Key)
Payment – PaymentID (Primary Key)
Delivery – DeliveryID (Primary Key)
Address – CustomerID + AddressID (Composite PK, Weak Entity)

IMPORTANT ATTRIBUTES

Customer: CustomerID, Name, Email, PhoneNo. (multi-valued)
Product: ProductID, ProductName, Price, Images (multi-valued)
Order: OrderID, OrderDate, TotalAmount
OrderItem: OrderID, ProductID, Quantity, UnitPrice, Discount
Seller: SellerID, SellerName, Email, PhoneNo., GSTIN
Category: CategoryID, CategoryName
Payment: PaymentID, Amount, PaymentMethod, PaymentDate
Delivery: DeliveryID, DeliveryDate, Status, CourierName, TrackingNo.
Address: AddressID, DoorNo, Street, City, State, Pincode

COMPOSITE ATTRIBUTE

AddressType can be represented as a composite attribute with Home, Work and Other.

RELATIONSHIPS AND CARDINALITIES

Customer PLACES Order → 1:N
Order HAS OrderItem → 1:N
OrderItem FOR Product → N:1
Customer HAS Address → 1:N
Product BELONGS TO Category → N:1
Seller SELLS Product → 1:N
Order HAS Payment → 1:1
Order DELIVERED BY Delivery → 1:1

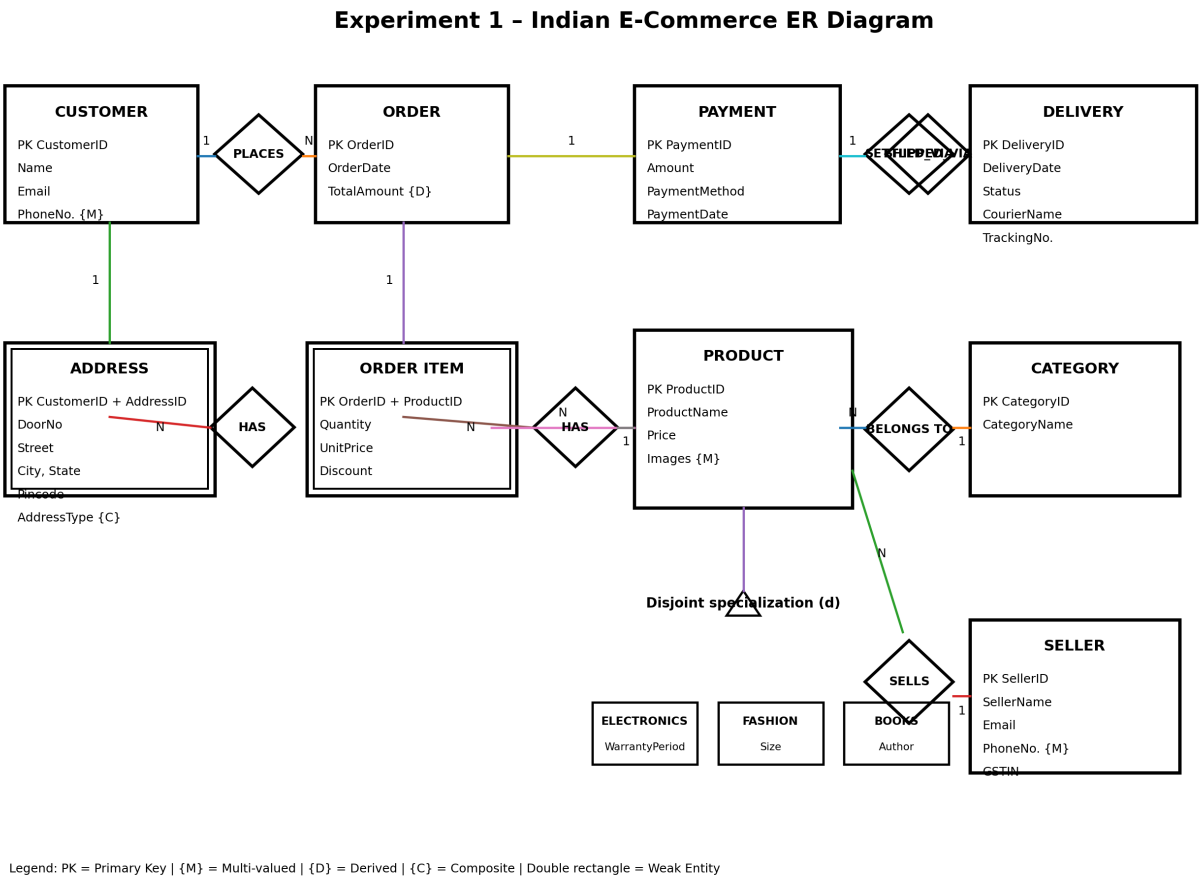
WEAK ENTITIES

OrderItem is a weak entity identified by OrderID + ProductID. Address can be represented as a weak entity identified by CustomerID + AddressID.

SPECIALIZATION

Product is specialized into Electronics (WarrantyPeriod), Fashion (Size), and Books (Author). The specialization is disjoint: one product belongs to only one subtype.

ER DIAGRAM



Legend: PK = Primary Key | {M} = Multi-valued | {D} = Derived | {C} = Composite | Double rectangle = Weak Entity

The diagram uses primary keys, weak entities, multi-valued attributes, the derived TotalAmount attribute, cardinalities, participation constraints, and disjoint Product specialization as specified in the source experiment.